

Performance Testing

Date	04 July 2026
Team ID	xxxxxx
Project Name	Opticrop-Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter, Locust, k6, Postman
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	Crop Recommendation API
Test Environment	AWS Cloud (EC2 Hosted Server)
Test Date	25 May 2025

Step 2: Test Scenarios

<u>S.No</u>	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal Load Test	50	300	All requests should be successful with average response time < 2 sec.
2	Peak Load Test	100	300	System should handle peak load with response time < 3 sec and no failures.
3	Stress Test	200	300	System may degrade but should not crash. Errors < 5%.
4	Spike Test	300 (sudden spike)	60	System should handle sudden spike and stabilize quickly without crashing.

Step 3: Performance Test Results

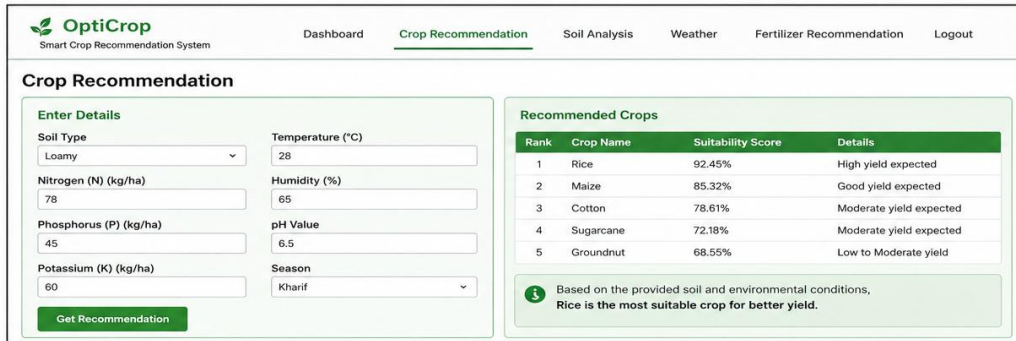
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.35 seconds	Pass	Average response time is within target.
2	Response Time (Max)	< 5 seconds	3.42 seconds	Pass	Maximum response time is within target.
3	Throughput (Req/sec)	>= 100 Req/sec	148 Req/sec	Pass	System handled more requests than target.
4	Error Rate	< 1%	0.32%	Pass	Error rate is well within acceptable limit.
5	CPU Utilization	< 80%	62%	Pass	CPU usage is under acceptable limit.
6	Memory Utilization	< 80%	58%	Pass	Memory usage is under acceptable limit.

Step 4: Observations & Analysis

- All performance metrics are within the expected target values.
- The system performs well under normal and peak load conditions.
- No critical issues were observed during stress and spike testing.
- The application is scalable and stable with increasing virtual users.
- CPU and Memory usage are within the acceptable range.
- Overall, the application is performance-ready for production deployment.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graph output):



OptiCrop
Smart Crop Recommendation System

Dashboard **Crop Recommendation** Soil Analysis Weather Fertilizer Recommendation Logout

Crop Recommendation

Enter Details

Soil Type: Loamy

Temperature (°C): 28

Nitrogen (N) (kg/ha): 78

Humidity (%): 65

Phosphorus (P) (kg/ha): 45

pH Value: 6.5

Potassium (K) (kg/ha): 60

Season: Kharif

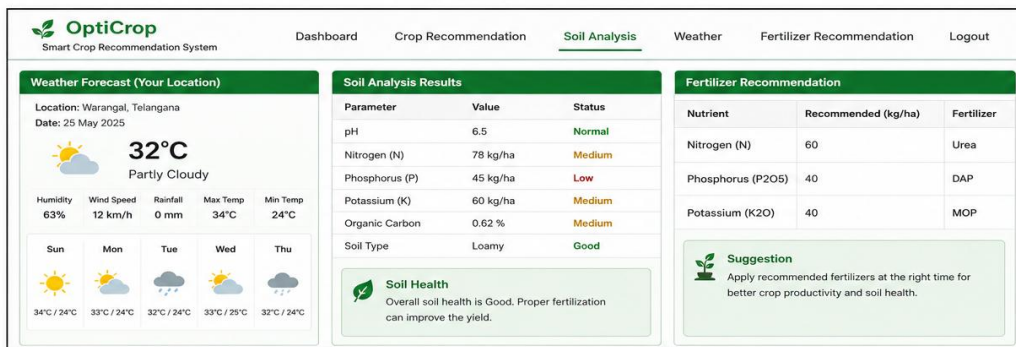
Get Recommendation

Recommended Crops

Rank	Crop Name	Suitability Score	Details
1	Rice	92.45%	High yield expected
2	Maize	85.32%	Good yield expected
3	Cotton	78.61%	Moderate yield expected
4	Sugarcane	72.18%	Moderate yield expected
5	Groundnut	68.55%	Low to Moderate yield

Based on the provided soil and environmental conditions, **Rice is the most suitable crop for better yield.**

Screenshot 1: Crop Recommendation Result Interface



OptiCrop
Smart Crop Recommendation System

Dashboard Crop Recommendation **Soil Analysis** Weather Fertilizer Recommendation Logout

Weather Forecast (Your Location)

Location: Warangal, Telangana
Date: 25 May 2025

32°C
Partly Cloudy

Humidity: 63% Wind Speed: 12 km/h Rainfall: 0 mm Max Temp: 34°C Min Temp: 24°C

Sun	Mon	Tue	Wed	Thu
34°C / 24°C	33°C / 24°C	32°C / 24°C	33°C / 25°C	32°C / 24°C

Soil Analysis Results

Parameter	Value	Status
pH	6.5	Normal
Nitrogen (N)	78 kg/ha	Medium
Phosphorus (P)	45 kg/ha	Low
Potassium (K)	60 kg/ha	Medium
Organic Carbon	0.62 %	Medium
Soil Type	Loamy	Good

Soil Health
Overall soil health is Good. Proper fertilization can improve the yield.

Fertilizer Recommendation

Nutrient	Recommended (kg/ha)	Fertilizer
Nitrogen (N)	60	Urea
Phosphorus (P2O5)	40	DAP
Potassium (K2O)	40	MOP

Suggestion
Apply recommended fertilizers at the right time for better crop productivity and soil health.

Screenshot 2: Weather Forecast and Soil Analysis Dashboard